Secure Backup and Recovery Management System

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a secure backup and recovery management system for a university that protects critical data from academic departments, administrative systems, research platforms, and shared services. The system should allow authorized administrators to define backup policies, initiate and monitor backup jobs, restore approved data, and review backup status across multiple campus locations. It should detect failed or incomplete backup operations, generate security and operational alerts, maintain audit logs of backup and restore activities, and restrict backup administration based on user roles. The platform should integrate with existing university storage and identity services and remain secure, reliable, scalable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Access restoration	Returning access that was previously restricted, within the source-defined permission boundary.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.
Policy	A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a secure backup and recovery management system for a university that protects critical data from academic departments, administrative systems, research platforms, and shared services.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- protect critical data from academic departments, administrative systems, research platforms, and shared services - FR-001
- allow authorized administrators to define backup policies - FR-002
- allow authorized administrators to initiate backup jobs - FR-003
- allow authorized administrators to monitor backup jobs - FR-004
- allow authorized administrators to restore approved data - FR-005
- allow authorized administrators to review backup status across multiple campus locations - FR-006
- detect failed or incomplete backup operations - FR-007
- generate security and operational alerts - FR-008
- maintain audit logs of backup and restore activities - FR-009
- restrict backup administration based on user roles - FR-010
- integrate with existing university storage and identity services - FR-011

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
User	user	Uses permitted functions; each source class keeps its specified permissions.
Admin	administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Which fields must the audit logs contain?
- What retention period is required for the audit logs?

- Which security acceptance outcomes must be demonstrated?
- What measurable reliability and recovery targets must the system meet?
- What measurable scalability targets must the system meet?
- What criteria define acceptable manageability?
- How are the stated roles and authorized permissions assigned?
- What conditions define the requested detections and their alert triggers?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-010. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

Related requirements: FR-011. Interface details beyond those requirements remain unspecified.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow authorized administrators to define backup policies.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall allow authorized administrators to initiate backup jobs.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall allow authorized administrators to monitor backup jobs.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall allow authorized administrators to restore approved data.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall allow authorized administrators to review backup status across multiple campus locations.

Traceability: E-002, TEST-006

FR-007 · Should · explicit

The system shall detect failed or incomplete backup operations.

Traceability: E-003, TEST-007

FR-008 · Should · explicit

The system shall generate security and operational alerts.

Traceability: E-003, TEST-008

FR-009 · Should · explicit

The system shall maintain audit logs of backup and restore activities.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall restrict backup administration based on user roles.

Traceability: E-003, TEST-010

FR-011 · Should · explicit

The system shall integrate with existing university storage and identity services.

Traceability: E-004, TEST-011

3.3 Performance requirements

NFR-004 · Should · explicit

The system shall remain scalable.

Traceability: E-004, TEST-015

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall remain reliable.

Traceability: E-004, TEST-014

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-008, FR-009, FR-010.

NFR-001 · Must · explicit

The system shall be secure.

Traceability: E-001, TEST-012

NFR-002 · Should · explicit

The system shall remain secure.

Traceability: E-004, TEST-013

3.6.4 Maintainability

NFR-005 · Should · explicit

The system shall remain manageable.

Traceability: E-004, TEST-016

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: User identities and authenticated access

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Unauthorized use of a legitimate identity could expose resources or alter system state.

Linked requirements: FR-011, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.2 THR-002: Role and permission boundaries

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: A lower-privilege user could access restricted functions or sensitive resources.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010

Original evidence: E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.3 THR-003: Administrative security decisions

Scenario: Derived hypothesis: A malicious or compromised administrator misuses a source-permitted management operation to weaken protection or disrupt legitimate use through the stated administrative workflow.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Misused administrative authority could undermine protection or interrupt legitimate use.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006

Original evidence: E-002

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.4 THR-004: Integration trust paths

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Untrusted integration data could cause incorrect decisions or hide security events.

Linked requirements: FR-011, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.5 THR-005: Monitoring and security alerts

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Relevant suspicious activity could remain undetected or receive delayed investigation.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010

Original evidence: E-002, E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

A.6 THR-006: Security and administrative audit evidence

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Investigators could lose reliable evidence of the events required by the brief.

Linked requirements: FR-007, FR-008, FR-009, FR-010

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

B.1 RISK-001

Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Proposed treatment: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-011, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.2 RISK-002

Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Proposed treatment: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010

Original evidence: E-002, E-003

B.3 RISK-003

Derived risk affecting Administrative security decisions: Misused administrative authority could undermine protection or interrupt legitimate use.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A malicious or compromised administrator misuses a source-permitted management operation to weaken protection or disrupt legitimate use through the stated administrative workflow.

Proposed treatment: Review misuse cases for each source-permitted administrative operation and compare its outcomes and audit coverage with the stated management boundary. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006

Original evidence: E-002

B.4 RISK-004

Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Proposed treatment: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-011, NFR-002, NFR-003, NFR-004, NFR-005

Original evidence: E-004

B.5 RISK-005

Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Proposed treatment: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010

Original evidence: E-002, E-003

B.6 RISK-006

Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Proposed treatment: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-007, FR-008, FR-009, FR-010

Original evidence: E-003

C SDLC recommendation and delivery workflow

Proposed lifecycle: Incremental

The Incremental model is selected based on derived evidence showing the need for integrated security review (E-002), operational validation through monitoring feedback (E-002, E-003), staged deployment across multiple locations (E-002), and integration verification of connected systems (E-004). These factors collectively support a phased, iterative approach that enables continuous assurance without assuming continuous deployment or full-scale rollout. The Incremental model scores highest at 66 points—outperforming Hybrid (65) and DevSecOps (63)—due to superior alignment with operational validation and deployment scope requirements. While security assurance is well-supported by all models, the derived operational feedback and staged rollout opportunities are most effectively addressed in an incremental lifecycle. The Hybrid model lacks sufficient weighting on deployment scope and integration assurance, while DevSecOps underweights deployment scope and overemphasizes security at the expense of operational validation. Spiral, Agile, and V-Model scores are significantly lower due to absence of differentiating evidence. Unknown factors such as budget, failure consequences, legacy systems, and verification methods remain unaddressed, making this recommendation provisional. The decision status explicitly notes that delivery constraints and team

capacity must be confirmed before adoption.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- security_assurance = integrated security review (derived). Security decisions benefit from security review and tests throughout delivery. Original evidence: E-002.
- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-002, E-003.
- deployment_scope = staged rollout opportunity (derived). Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. Original evidence: E-002.
- integration_assurance = integration validation (derived). Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. Original evidence: E-004.

Candidate	Suitability points
Incremental	66
Hybrid	65
DevSecOps	63
Spiral	51
Agile	40
V-Model	40
Waterfall	40

C.1.1 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003)
- Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002)
- Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.2 Hybrid suitability reasons

- Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003)
- Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002)
- Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.3 DevSecOps suitability reasons

- Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-002, E-003)
- Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-002)

C.1.4 Spiral suitability reasons

- Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002)
- Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

C.1.5 Agile suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.; FR-002: The system shall allow authorized administrators to define backup policies.; FR-003: The system shall allow authorized administrators to initiate backup jobs.; FR-004: The system shall allow authorized administrators to monitor backup jobs.; FR-005: The system shall allow authorized administrators to restore approved data.; FR-006: The system shall allow authorized administrators to review backup status across multiple campus locations.; FR-007: The system shall detect failed or incomplete backup operations.; FR-008: The system shall generate security and operational alerts.; FR-009: The system shall maintain audit logs of backup and restore activities.; FR-010: The system shall restrict backup administration based on user roles.; FR-011: The system shall integrate with existing university storage and identity services.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.2 Incremental delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall protect critical data from academic departments, administrative systems, research platforms, and shared services.; FR-002: The system shall allow authorized administrators to define backup policies.; FR-003: The system shall allow authorized administrators to initiate backup jobs.; FR-004: The system shall allow authorized administrators to monitor backup jobs.; FR-005: The system shall allow authorized administrators to restore approved data.; FR-006: The system shall allow authorized administrators to review backup status across multiple campus locations.; FR-007: The system shall detect failed or incomplete backup operations.; FR-008: The system shall generate security and operational alerts.; FR-009: The system shall maintain audit logs of backup and restore activities.; FR-010: The system shall restrict backup administration based on user roles.; FR-011: The system shall integrate with existing university storage and identity services.; NFR-001: The system shall be secure.;

NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.

- Activity: In an authorized isolated test environment, execute TEST-001: Given: the system protects critical data from academic departments, administrative systems, research platforms, and shared services When: the system initiates a protection operation for designated data assets Expected: the protected data remains accessible and intact without unauthorized modification or loss
- Activity: In an authorized isolated test environment, execute TEST-002: Given: an administrator defines a backup policy for a data set When: the administrator initiates the policy definition operation Expected: the named policy is available for examination in the system interface
- Activity: In an authorized isolated test environment, execute TEST-003: Given: an administrator initiates a backup job for a data set When: the administrator starts the backup job operation Expected: the named job enters its active state and is visible in the system interface
- Activity: In an authorized isolated test environment, execute TEST-004: Given: an administrator monitors a running backup job When: the administrator performs a monitoring action on the job Expected: the system returns status updates for the named job
- Activity: In an authorized isolated test environment, execute TEST-005: Given: an administrator restores a data set from a backup copy When: the administrator executes the restore operation Expected: the named data or configuration is restored to its expected state and is accessible in the system interface
- Activity: In an authorized isolated test environment, execute TEST-006: Given: an administrator requests backup status for multiple campus locations When: the administrator performs a cross-location status query Expected: the named items are returned in the requested format and are available for examination
- Activity: In an authorized isolated test environment, execute TEST-007: Given: system detects a failed or incomplete backup operation When: the system initiates detection of a failed or incomplete backup operation Expected: the system reports the failure with identifiable metadata
- Activity: In an authorized isolated test environment, execute TEST-008: Given: system identifies an alert condition for security or operational risk When: the system initiates the alert condition Expected: the requested alert exists in the designated channel
- Activity: In an authorized isolated test environment, execute TEST-009: Given: system records backup and restore activities When: a backup or restore operation is completed Expected: The system records backup and restore activities when a backup or restore operation is completed, as evidenced by external test capture of the inspection record.
- Activity: In an authorized isolated test environment, execute TEST-010: Given: user attempts to perform backup administration When: the user initiates an administrative action Expected: the system denies access if the user role lacks backup administration privileges
- Activity: In an authorized isolated test environment, execute TEST-011: Given: system integrates with existing university storage and identity services using a controlled exchange When: the system initiates a connection request to the external service Expected: the external service responds with an authenticated session token and confirms successful path establishment
- Activity: In an authorized isolated test environment, execute TEST-012: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD/uncertain. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific metrics or thresholds for security assurance are provided in the current draft.
- Activity: In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall remain secure during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes

measurable targets for security assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.

- Activity: In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for this quality goal: the system shall remain reliable during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for reliability assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
- Activity: In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for this quality goal: the system shall remain scalable during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for scalability assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
- Activity: In an authorized isolated test environment, execute TEST-016: Given: A draft acceptance plan for this quality goal: the system shall remain manageable during backup and recovery operations. When: An analyst inspects or reviews that acceptance plan to determine whether it includes measurable targets for manageability assurance. Expected: The plan's measurable targets remain TBD or unconfirmed; no specific metric, threshold, or test case is defined in the current draft.
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

C.2.3 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, FR-011, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package

- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review implementation and acceptance evidence against FR-008: The system shall generate security and operational alerts.
- Security check: Review implementation and acceptance evidence against NFR-001: The system shall be secure.
- Security check: Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

Canonical content SHA-256: c9d3fb77085f6f067c8cc4b0e60b6731b45ba222e38d9cb7cf774da63250bae3